THE CHINESE REMAINDER THEOREM

KEITH CONRAD

We should thank the Chinese for their wonderful remainder theorem.
Glenn Stevens

1. Introduction

The Chinese remainder theorem says we can uniquely solve every pair of congruences having relatively prime moduli.

Theorem 1.1. *Let m and n be relatively prime positive integers. For all integers a and b, the pair of congruences*

$$x \equiv a \bmod m, \quad x \equiv b \bmod n$$

has a solution, and this solution is uniquely determined modulo mn.

What is important here is that m and n are relatively prime. There are *no* constraints at all on a and b.

Example 1.2. The congruences $x \equiv 6 \bmod 9$ and $x \equiv 4 \bmod 11$ hold when $x = 15$, and more generally when $x \equiv 15 \bmod 99$, and they do not hold for other x. The modulus 99 is $9 \cdot 11$.

We will prove the Chinese remainder theorem, including a version for more than two moduli, and see some ways it is applied to study congruences.

2. A proof of the Chinese remainder theorem

Proof. First we show there is always a solution. Then we will show it is unique modulo mn.

Existence of Solution. To show that the simultaneous congruences

$$x \equiv a \bmod m, \quad x \equiv b \bmod n$$

have a common solution in $\mathbf{Z}$, we give two proofs.

First proof: Write the first congruence as an equation in $\mathbf{Z}$, say $x = a + my$ for some $y \in \mathbf{Z}$. Then the second congruence is the same as

$$a + my \equiv b \bmod n.$$

Subtracting a from both sides, we need to solve for y in

$$my \equiv b - a \bmod n. \tag{2.1}$$

Since $(m, n) = 1$, we know $m \bmod n$ is invertible. Let m' be an inverse for $m \bmod n$, so $mm' \equiv 1 \bmod n$. Multiplying through (2.1) by m', we have $y \equiv m'(b - a) \bmod n$, so $y \equiv m'(b - a) + nz$ where $z \in \mathbf{Z}$. Then

$$x = a + my = a + m(m'(b - a) + nz) = a + mm'(b - a) + mnz.$$

So *if* x satisfies the original two congruences it must have this form. Let's now check this expression, for every $z \in \mathbf{Z}$, really satisfies the original two congruences:

$$a + mm'(b-a) + mnz \equiv a + 0 + 0 \equiv a \bmod m$$

and

$$a + mm'(b-a) + mnz \equiv a + 1(b-a) + 0 \equiv b \bmod n.$$

Second proof: Write both congruences as equations in $\mathbf{Z}$: $x = a + my$ and $x = b + nz$ for integers y and z that need to be determined. (Why would it be a bad idea to write $x = a + my$ and $x = b + ny$?) The integers of the form $a + my$ are the numbers that are congruent to $a \bmod m$, and the integers of the form $b + nz$ are the numbers that are congruent to $b \bmod n$. Finding a common solution to the two congruences amounts to finding y and z in $\mathbf{Z}$ such that

$$a + my = b + nz,$$

which is the same as

$$my - nz = b - a. \tag{2.2}$$

Can we find such y and z for all a, b, m, and n where $(m, n) = 1$? Bezout's identity tells us 1 is a $\mathbf{Z}$-linear combination of m and n, and therefore every integer is a $\mathbf{Z}$-linear combination of m and n (why?). Therefore integers y and z satisfying (2.2) exist.

Uniqueness of Solution. If $x = c$ and $x = c'$ both satisfy

$$x \equiv a \bmod m, \quad x \equiv b \bmod n,$$

then we have $c \equiv c' \bmod m$ and $c \equiv c' \bmod n$. Then $m \mid (c - c')$ and $n \mid (c - c')$. Since $(m, n) = 1$, the product mn divides $c - c'$, which means $c \equiv c' \bmod mn$. This shows all solutions to the initial pair of congruences are the same modulo mn. □

3. Extension to more than two congruences

The Chinese remainder theorem can be extended from two congruences to an arbitrary finite number of congruences, but we have to be careful about the way in which the moduli are relatively prime. Consider the three congruences

$$x \equiv 1 \bmod 6, \quad x \equiv 4 \bmod 10, \quad x \equiv 7 \bmod 15.$$

While there is no common factor of 6, 10, and 15 greater than 1, these congruences do not admit a common solution: every solution to the first congruence is odd, while every solution to the second congruence is even. When we have more than two moduli, we have to be sensitive to the difference between saying the moduli are collectively relatively prime (no common factor greater than 1 divides all the moduli) and pairwise relatively prime (each pair of moduli have no common factor greater than 1). For instance, 6, 10, and 15 are collectively relatively prime but not pairwise relatively prime. Here is a more general form of the Chinese remainder theorem that allows more than two moduli.

Theorem 3.1. *For $r \geq 2$, let $m_1, m_2, \ldots, m_r$ be nonzero integers that are pairwise relatively prime: $(m_i, m_j) = 1$ for $i \neq j$. Then, for all integers $a_1, a_2, \ldots, a_r$, the system of congruences*

$$x \equiv a_1 \bmod m_1, \quad x \equiv a_2 \bmod m_2, \quad \ldots, \quad x \equiv a_r \bmod m_r,$$

has a solution, and this solution is uniquely determined modulo $m_1 m_2 \cdots m_r$.

Example 3.2. The congruences $x \equiv 1 \bmod 3$, $x \equiv 2 \bmod 5$, $x \equiv 2 \bmod 7$ are satisfied when $x = 37$, more generally for all $x \equiv 37 \bmod 105$ and for no other x. Note $105 = 3 \cdot 5 \cdot 7$.

Proof. First we show there is always a solution. Then we will show it is unique modulo $m_1 m_2 \cdots m_r$.

Existence of Solution. We argue by induction on r. The base case $r = 2$ is Theorem 1.1, which has been proved already.

Now we pass to the inductive step. Suppose all simultaneous congruences with r pairwise relatively prime moduli can be solved. Consider a system of simultaneous congruences with $r+1$ pairwise relatively prime moduli:

$$x \equiv a_1 \bmod m_1, \quad \ldots, \quad x \equiv a_r \bmod m_r, \quad x \equiv a_{r+1} \bmod m_{r+1},$$

where $(m_i, m_j) = 1$ for all $i \neq j$ and the a_i's are arbitrary. By the inductive hypothesis, there is a solution b to the first r congruences, say

$$b \equiv a_1 \bmod m_1, \quad b \equiv a_2 \bmod m_2, \quad \ldots, \quad b \equiv a_r \bmod m_r.$$

Now consider the system of *two* congruences

$$x \equiv b \bmod m_1 m_2 \cdots m_r, \quad x \equiv a_{r+1} \bmod m_{r+1}. \tag{3.1}$$

Since $(m_i, m_{r+1}) = 1$ for $i = 1, 2, \ldots, r$, we have $(m_1 m_2 \cdots m_r, m_{r+1}) = 1$, so the two moduli in (3.1) are relatively prime. Then by the case of two congruences, namely Theorem 1.1, there is a solution to (3.1). Call it c. Since $c \equiv b \bmod m_1 m_2 \cdots m_r$, we have $c \equiv b \bmod m_i$ for $i = 1, 2, \ldots, r$. From the choice of b we have $b \equiv a_i \bmod m_i$ for $i = 1, 2, \ldots, r$. Therefore $c \equiv a_i \bmod m_i$ for $i = 1, 2, \ldots, r$. Also, $c \equiv a_{r+1} \bmod m_{r+1}$ from the choice of c, so we see c satisfies the $r+1$ given congruences.

This concludes the inductive step, so a solution exists.

Uniqueness of Solution. If $x = c$ and $x = c'$ both satisfy

$$x \equiv a_1 \bmod m_1, \quad x \equiv a_2 \bmod m_2, \quad \ldots, \quad x \equiv a_r \bmod m_r,$$

then we have $c \equiv c' \bmod m_i$ for $i = 1, 2, \ldots, r$, so $m_i \mid (c - c')$ for $i = 1, 2, \ldots, r$. Since the m_i's are pairwise relatively prime, their product $m_1 m_2 \cdots m_r$ divides $c - c'$, which means $c \equiv c' \bmod m_1 m_2 \cdots m_r$. This shows all solutions to the given system of congruences are the same when viewed modulo $m_1 m_2 \cdots m_r$. □

4. Applications

The significance of the Chinese remainder theorem is that it often reduces a question about modulus mn, where $(m, n) = 1$, to the same question for modulus m and n separately. In this way, questions about modular arithmetic can often be reduced to the special case of prime power moduli. We will see how this works for several counting problems, often using two features of modular arithmetic with two moduli:[1]

- if $d \mid m$ it makes sense to reduce integers mod m to integers mod d: if $x \equiv y \bmod m$ then $x \equiv y \bmod d$. For example, if $x \equiv y \bmod 10$ then $x \equiv y \bmod 5$ since if $x - y$ is divisible by 10 then it is also divisible by 5. (In contrast, it makes no sense to reduce x mod 10 to x mod 3, since there are congruent numbers mod 10 that are incongruent mod 3, such as 1 and 11.)
- if $x \equiv y \bmod m$ and $x \equiv y \bmod n$ and $(m, n) = 1$ then $x \equiv y \bmod mn$. This was used in the uniqueness part of the proof of the Chinese remainder theorem.

Our first application is to counting units mod m.

[1]The Chinese remainder theorem for two or more moduli that need not be prime powers has real-world uses, such as determining a target range using multiple radar signals [1, Sec.10.5, 10.6].

Theorem 4.1. *For relatively prime positive integers m and n, $\varphi(mn) = \varphi(m)\varphi(n)$.*

Proof. We work with the sets

$$U_m = \{a \bmod m, (a,m) = 1\}, \quad U_n = \{b \bmod n, (b,n) = 1\},$$
$$U_{mn} = \{c \bmod mn, (c,mn) = 1\}.$$

Then $|U_m| = \varphi(m)$, $|U_n| = \varphi(n)$, and $|U_{mn}| = \varphi(mn)$. To show $\varphi(mn) = \varphi(m)\varphi(n)$, we will write down a bijection between U_{mn} and $U_m \times U_n$, which implies the two sets have the same size, and that is what the theorem is saying (since $|U_m \times U_n| = \varphi(m)\varphi(n)$).

Let $f\colon U_{mn} \to U_m \times U_n$ by the rule

$$f(c \bmod mn) = (c \bmod m, c \bmod n).$$

For $c \in U_{mn}$, we have $(c, mn) = 1$, so (c,m) and (c,n) equal 1, so $c \bmod m$ and $c \bmod n$ are units. Let's stop for a moment to take a look at an example of this function.

Take $m = 3$ and $n = 5$: $U_3 = \{1,2\}$, $U_5 = \{1,2,3,4\}$, and $U_{15} = \{1,2,4,7,8,11,13,14\}$. The following table shows the values of the function f on each number in U_{15}. Notice that the values fill up all of $U_3 \times U_5$ without repetition.

$c \bmod 15$	$f(c \bmod 15)$
1	$(1,1)$
2	$(2,2)$
4	$(4,4) = (1,4)$
7	$(7,7) = (1,2)$
8	$(8,8) = (2,3)$
11	$(11,11) = (2,1)$
13	$(13,13) = (1,3)$
14	$(14,14) = (2,4)$

There are 2 units modulo 3 and 4 units modulo 5, leading to 8 ordered pairs of units modulo 3 and units modulo 5: (1,1), (1,2), (1,3), (1,4), (2,1), (2,2), (2,3), and (2,4). All these pairs show up (and just once) in the second column of the table.

We return to the general situation and show $f\colon U_{mn} \to U_m \times U_n$ is a bijection.

To see that f is one-to-one, suppose $f(k \bmod mn) = f(\ell \bmod mn)$. Then $k \equiv \ell \bmod m$ and $k \equiv \ell \bmod n$, so since $(m,n) = 1$ (aha!), we have $k \equiv \ell \bmod mn$. That means $k = \ell$ in U_{mn}, so f is one-to-one.

Now we show f is onto. Pick a pair $(a \bmod m, b \bmod n) \in U_m \times U_n$. By the Chinese remainder theorem we can solve $c \equiv a \bmod m$ and $c \equiv b \bmod n$ for a $c \in \mathbf{Z}$. Is $(c, mn) = 1$? Since $a \bmod m$ is a unit and $c \equiv a \bmod m$, $c \bmod m$ is a unit so $(c,m) = 1$. Since $b \bmod n$ is a unit and $c \equiv b \bmod n$, $c \bmod n$ is a unit so $(c,n) = 1$. From $(c,m) = 1$ and $(c,n) = 1$ we get $(c, mn) = 1$, so $c \in U_{mn}$. From the congruence conditions on c, we have $f(c) = (a,b)$. $\square$

Corollary 4.2. *For a positive integer m,*

$$\varphi(m) = m \prod_{p \mid m} \left(1 - \frac{1}{p}\right),$$

where the product runs over the primes p dividing m.

Proof. The formula is clear for $m = 1$ (interpreting an empty product as 1).

Now suppose $m > 1$, and factor m into prime powers:

$$m = p_1^{e_1} p_2^{e_2} \cdots p_r^{e_r}.$$

The $p_i^{e_i}$'s are pairwise relatively prime. By an extension of Theorem 4.1 from two relatively prime terms to an arbitrary number of pairwise relatively prime terms (just induct on the number of terms), we have

$$\varphi(m) = \varphi(p_1^{e_1})\varphi(p_2^{e_2})\cdots\varphi(p_r^{e_r}).$$

Now using the formula for φ on prime powers,

$$\begin{aligned}\varphi(m) &= p_1^{e_1-1}(p_1-1)p_2^{e_2-1}(p_2-1)\cdots p_r^{e_r-1}(p_r-1)\\ &= p_1^{e_1}\left(1-\frac{1}{p_1}\right)p_2^{e_2}\left(1-\frac{1}{p_2}\right)\cdots p_r^{e_r}\left(1-\frac{1}{p_r}\right)\\ &= m\prod_{p\mid m}\left(1-\frac{1}{p}\right).\end{aligned}$$

□

Example 4.3. To compute $\varphi(540) = \varphi(2^2\cdot 3^3\cdot 5)$, we have

$$\begin{aligned}\varphi(540) &= 540\left(1-\frac{1}{2}\right)\left(1-\frac{1}{3}\right)\left(1-\frac{1}{5}\right)\\ &= 540\cdot\frac{1}{2}\cdot\frac{2}{3}\cdot\frac{4}{5}\\ &= 18\cdot 8\\ &= 144.\end{aligned}$$

An alternate calculation is

$$\begin{aligned}\varphi(540) &= \varphi(4)\varphi(27)\varphi(5)\\ &= (4-2)(27-9)(5-1)\\ &= 2\cdot 18\cdot 4\\ &= 144.\end{aligned}$$

Example 4.4. A table of values of $\varphi(m)$, as shown below, suggests that $\varphi(m)$ is even when $m > 2$, which is true. But not every positive even number is a value of the φ-function.

m	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
$\varphi(m)$	1	1	2	2	4	2	6	4	6	4	10	4	12	6	8	8	16

We will use the general formula for $\varphi(m)$ to show for all odd primes $q > 3$ that there is no solution to $\varphi(m) = 2q^2$. (For example, taking $q = 5, 7$, and 11, $\varphi(m)$ is never 50, 98, or 242. But $2\cdot 3^2 = \varphi(3^3)$.) For each odd prime factor p of m, $\varphi(m)$ is divisible by $p-1$, which is even, so if m has more than 2 odd prime factors then $\varphi(m)$ would be divisible by 4. Thus when $\varphi(m) = 2q^2$, m has at most one odd prime factor, so $m = 2^e$, $2^e p^f$, or p^f (all exponents are positive).

- If $m = 2^e$ then $\varphi(m) = 2^{e-1}$ is a power of 2, which is not true.
- If $m = p^f$ then $\varphi(m) = p^{f-1}(p-1)$, so if this is $2q^2$ then $p^{f-1}(p-1)/2 = q^2$. If $f \geq 2$ then $p \mid q^2$, so $p = q$ and $q^{f-1}(q-1)/2 = q^2$. Every prime factor of $(q-1)/2$ has to be q, which is impossible, so $(q-1)/2 = 1$ and thus $q = 3$, so when $q > 3$ we can't have $m = p^f$.
- If $m = 2^e p^f$ then $\varphi(m) = \varphi(2^e)\varphi(p^f) = 2^{e-1}(p-1)p^{f-1}$. The factor $p-1$ is even, and if $e \geq 2$ then 2^{e-1} is also even, which contradicts $\varphi(m) = 2q^2$. Therefore $e = 1$,

so $m = 2p^f$ and $\varphi(m) = (p-1)p^{f-1} = \varphi(p^f)$. By the previous case, if $\varphi(p^f) = 2q^2$ then $q = 3$.

We now leave units mod m and look at squares mod m.

Theorem 4.5. *For $m \in \mathbf{Z}^+$ with $m \geq 2$, let $S_m = \{x^2 \bmod m\}$ be the set of squares modulo m. When $(m,n) = 1$, $|S_{mn}| = |S_m| \cdot |S_n|$.*

Note S_m is *all* squares modulo m, including 0. So $S_5 = \{0,1,4\}$, for example.

Proof. We will use the Chinese remainder theorem *twice*.

If $a \equiv x^2 \bmod mn$ then $a \equiv x^2 \bmod m$ and $a \equiv x^2 \bmod n$. Thus a square modulo mn reduces to a square modulo m and a square modulo n. So we have a function $f\colon S_{mn} \to S_m \times S_n$ by $f(a \bmod mn) = (a \bmod m, a \bmod n)$. Let's take a look at an example.

Set $m = 3$ and $n = 5$, so $S_3 = \{0,1\}$, $S_5 = \{0,1,4\}$ and $S_{15} = \{0,1,4,6,9,10\}$. The table below gives the values of f on S_{15}. The values fill up $S_3 \times S_5$ without repetition.

$c \bmod 15$	$f(c \bmod 15)$
0	$(0,0)$
1	$(1,1)$
4	$(4,4) = (1,4)$
6	$(6,6) = (0,1)$
9	$(9,9) = (0,4)$
10	$(10,10) = (1,0)$

Returning to the general case, to show f is one-to-one let's suppose $f(c \bmod mn) = f(c' \bmod mn)$. Then $c \equiv c' \bmod m$ and $c \equiv c' \bmod n$, so $c \equiv c' \bmod mn$ since $(m,n) = 1$.

To show f is onto, pick a pair of squares $b \bmod m$ and $c \bmod n$, say $b \equiv y^2 \bmod m$ and $c \equiv z^2 \bmod n$. By the Chinese remainder theorem, there is an integer a satisfying

$$a \equiv b \bmod m, \quad a \equiv c \bmod n.$$

We want to say $f(a) = (b,c)$, but is $a \bmod mn$ a square? From the expressions for $b \bmod m$ and $c \bmod n$ as squares, $a \equiv y^2 \bmod m$ and $a \equiv z^2 \bmod n$, but y and z are not related to each other. They certainly don't have to be the same integer, so these two congruences on their own don't tell us $a \bmod mn$ is a square. Using the Chinese remainder theorem *again*, however, there is $x \in \mathbf{Z}$ such that

$$x \equiv y \bmod m, \quad x \equiv z \bmod n,$$

so $x^2 \equiv y^2 \bmod m$ and $x^2 \equiv z^2 \bmod n$. Therefore $a \equiv x^2 \bmod m$ and $a \equiv x^2 \bmod n$, so $a \equiv x^2 \bmod mn$, so $a \bmod mn$ is in fact a square. Thus $a \in S_{mn}$ and $f(a) = (b,c)$. □

Example 4.6. For a prime p, the number of nonzero squares mod p is $(p-1)/2$ and 0 is a square, so the total number of squares mod p is $1 + (p-1)/2 = (p+1)/2$. Thus $|S_p| = (p+1)/2$. So if $n = p_1 p_2 \dots p_r$ is squarefree, $|S_n| = |S_{p_1}| \cdots |S_{p_r}| = \frac{p_1+1}{2} \cdots \frac{p_r+1}{2}$. If $n = p_1^{e_1} \cdots p_r^{e_r}$, we have $|S_n| = |S_{p_1^{e_1}}| \cdots |S_{p_r^{e_r}}|$, so a formula for $|S_{p^e}|$ when $e > 1$ (which we don't give here) would lead to a formula for $|S_m|$ in general.

We turn now from counting all the squares mod m to counting how often something is a square mod m.

Example 4.7. We can write $1 \bmod 15$ as a square in *four* ways: $1 \equiv 1^2 \equiv 4^2 \equiv 9^2 \equiv 14^2 \bmod 15$.

Theorem 4.8. *Let $m \in \mathbf{Z}^+$ have prime factorization $p_1^{e_1} \cdots p_r^{e_r}$. For $a \in \mathbf{Z}$, the congruence $x^2 \equiv a \bmod m$ is solvable if and only if the separate congruences $x^2 \equiv a \bmod p_i^{e_i}$ are solvable for $i = 1, 2, \ldots, r$.*

Furthermore, if the congruence $x^2 \equiv a \bmod p_i^{e_i}$ has N_i solutions, then the congruence $x^2 \equiv a \bmod m$ has $N_1 N_2 \cdots N_r$ solutions.

Example 4.9. The congruences $x^2 \equiv 1 \bmod 3$ and $x^2 \equiv 1 \bmod 5$ each have two solutions, so $x^2 \equiv 1 \bmod 15$ has $2 \cdot 2 = 4$ solutions; we saw the four square roots of 1 mod 15 before the statement of Theorem 4.8.

Proof. If $x \in \mathbf{Z}$ satisfies $x^2 \equiv a \bmod m$, then $x^2 \equiv a \bmod p_i^{e_i}$ for all i.

Conversely, suppose each of the congruences $x^2 \equiv a \bmod p_i^{e_i}$ has a solution, say $x_i^2 \equiv a \bmod p_i^{e_i}$ for some integers x_i. Since the $p_i^{e_i}$'s are pairwise relatively prime, the Chinese remainder theorem tells us there is an x such that $x \equiv x_i \bmod p_i^{e_i}$ for all i. Then $x^2 \equiv x_i^2 \bmod p_i^{e_i}$ for all i, so $x^2 \equiv a \bmod p_i^{e_i}$ for all i. Since $x^2 - a$ is divisible by each $p_i^{e_i}$ it is divisible by m, so $x^2 \equiv a \bmod m$.

To count the solutions modulo m, we again use the Chinese remainder theorem. Any choice of solution $x_i \bmod p_i^{e_i}$ for each i fits together in exactly one way to a number $x \bmod m$, and this number will satisfy $x^2 \equiv a \bmod m$. Therefore we can count solutions modulo m by counting solutions modulo each $p_i^{e_i}$ and multiply the counts thanks to the independence of the choice of solutions for different primes. □

Example 4.10. To decide if 61 is a square modulo 75, we check whether 61 is a square modulo 3 and modulo 25. Since $61 \equiv 1 \bmod 3$, it is a square modulo 3. Since $61 \equiv 11 \equiv 6^2 \bmod 25$, it is a square modulo 25. Therefore 63 is a square modulo 75. In fact, we can get a square root by solving the congruences

$$x \equiv 1 \bmod 3, \quad x \equiv 6 \bmod 25.$$

A solution is $x = 31$, so $61 \equiv 31^2 \bmod 75$.

Remark 4.11. It is crucial to remember that using the Chinese remainder theorem requires the moduli to be relatively prime (more precisely, pairwise relatively prime). For a prime p you may be able to prove results about modulus p^{k+1} from similar results for moduli p and p^k, but the proof won't be based on the Chinese remainder theorem since p and p^k are not relatively prime. An example of such a result is that if p is an odd prime and $a \not\equiv 0 \bmod p$, then for $k \geq 2$, $a \bmod p^k$ is a square if and only if $a \bmod p$ is a square. Proving that has *nothing* to do with the Chinese remainder theorem.

If you scrutinize the proofs of Theorems 4.5 and 4.8 to see how it was important we were working with squares, you'll see that what really matters is that squaring is a polynomial expression. With this in mind, we get the following two generalizations from squares to values of other polynomials.

Theorem 4.12. *Let $f(x)$ be a polynomial with integer coefficients. For a positive integer $m \geq 2$, let $N_f(m) = |\{f(x) \bmod m : 0 \leq x \leq m-1\}|$ be the number of values of f on different integers mod m. If m has prime factorization*

$$m = p_1^{e_1} \cdots p_r^{e_r},$$

then $N_f(m) = N_f(p_1^{e_1}) \cdots N_f(p_r^{e_r})$.

Proof. Proceed as in the proof of Theorem 4.5, which is the special case $f(x) = x^2$. □

Theorem 4.13. *Let $f(x)$ be a polynomial with integer coefficients. For a positive integer m with prime factorization*

$$m = p_1^{e_1} \cdots p_r^{e_r},$$

the congruence $f(x) \equiv 0 \bmod m$ is solvable if and only if the congruences $f(x) \equiv 0 \bmod p_i^{e_i}$ are each solvable.

Moreover, if $f(x) \equiv 0 \bmod p_i^{e_i}$ has N_i solutions, then the congruence $f(x) \equiv 0 \bmod m$ has $N_1 N_2 \cdots N_r$ solutions.

Proof. Argue as in the proof of Theorem 4.8, which is the special case $f(x) = x^2 - a$. □

Theorem 4.13 tells us that finding solutions to a polynomial equation modulo positive integers is reduced by the Chinese remainder theorem to the case of understanding solutions modulo prime powers.

Consider now the following situation: $f(x)$ is a polynomial with integral coefficients and every value $f(n)$, for $n \in \mathbf{Z}$, is either a multiple of 2 or a multiple of 3. For instance, if $f(x) = x^2 - x$ then $f(n) = n^2 - n$ is even for all n. Or if $f(x) = x^3 - x$ then $f(n) = n^3 - n$ is a multiple of 3 for all n. But these examples are kind of weak: what about a mixed example where every $f(n)$ is a multiple of 2 or 3 but some $f(n)$ are multiples of 2 and not 3 while other $f(n)$ are multiples of 3 and not 2? Actually, no such polynomial exists! The only way $f(n)$ can be divisible either by 2 or 3 for all n is if it is a multiple of 2 for all n or a multiple of 3 for all n. To explain this, we will use the Chinese remainder theorem.

Theorem 4.14. *Let $f(x)$ be a polynomial with integral coefficients. Suppose for a finite set of primes $p_1, \ldots, p_r$ each $f(n)$ for $n \in \mathbf{Z}$ is divisible by some p_i. Then there is a p_i such that each $f(n)$ is divisible by that p_i.*

Proof. Suppose the conclusion is false. Then, for each p_i, there is an $a_i \in \mathbf{Z}$ such that p_i does not divide $f(a_i)$. Said differently, $f(a_i) \not\equiv 0 \bmod p_i$.

Since the p_i's for $i = 1, \ldots, r$ are different primes, we can use the Chinese remainder theorem to find an integer a such that $a \equiv a_i \bmod p_i$ for $i = 1, 2, \ldots, r$. Then $f(a) \equiv f(a_i) \bmod p_i$ for $i = 1, 2, \ldots, r$ (why?), so $f(a) \not\equiv 0 \bmod p_i$ for all i. However, the assumption in the theorem was that every value of the polynomial on integers is divisible by some p_i, so we have a contradiction. □

Remark 4.15. If $f(x)$ is a polynomial with integral coefficients and there is a finite set of nonzero pairwise relatively prime integers $m_1, \ldots, m_r$ such that, for every integer n, $f(n)$ is divisible by some m_i, then there is some m_i such that $m_i \mid f(n)$ for every $n \in \mathbf{Z}$. The proof of Theorem 4.14 carries over directly to this case, with m_i replacing p_i.

Our next application of the Chinese remainder theorem will describe examples where a polynomial equation can have solutions mod m for *all* $m \geq 2$ even without having integer solutions, provided it has two "well-chosen" rational solutions. The idea is best explained by an example.

Example 4.16. The equation $2x^2 + 7y^2 = 1$ obviously has no solution in $\mathbf{Z}$. It does have solutions in $\mathbf{Q}$, such as $(x, y) = (1/3, 1/3)$ and $(3/5, 1/5)$. We will use these two rational solutions to show the congruence $2x^2 + 7y^2 \equiv 1 \bmod m$ is solvable for every $m \geq 2$. The key point is that the two pairs of rational solutions have relatively prime denominators, 3 and 5.

Write out the equations with their rational solutions and clear denominators:

$$2\left(\frac{1}{3}\right)^2 + 7\left(\frac{1}{3}\right)^2 = 1 \Longrightarrow 2 \cdot 1^2 + 7 \cdot 1^2 = 3^2,$$
$$2\left(\frac{3}{5}\right)^2 + 7\left(\frac{1}{5}\right)^2 = 1 \Longrightarrow 2 \cdot 3^2 + 7 \cdot 1^2 = 5^2.$$

For $m \geq 2$, we can reduce both equations in integers modulo m:

$$2 \cdot 1^2 + 7 \cdot 1^2 \equiv 3^2 \bmod m, \quad 2 \cdot 3^2 + 7 \cdot 1^2 \equiv 5^2 \bmod m. \tag{4.1}$$

We will consider these congruence separately, then together.

Case 1: m is not divisible by 3. We know 3 mod m has an inverse, say $3d \equiv 1 \bmod m$. Morally, d is "1/3 mod m". Multiplying through the first congruence in (4.1) by d^2:

$$2 \cdot 1^2 + 7 \cdot 1^2 \equiv 3^2 \bmod m \Rightarrow 2d^2 + 7d^2 \equiv 1 \bmod m, \tag{4.2}$$

and the solution (d, d) mod m should be regarded as "$(1/3, 1/3)$ mod m."

Case 2: m is not divisible by 5. Some integer d' satisfies $5d' \equiv 1 \bmod m$ and multiplying through the second congruence in (4.1) by d'^2 gives us

$$2 \cdot 3^2 + 7 \cdot 1^2 \equiv 5^2 \bmod m \Rightarrow 2(3d')^2 + 7d'^2 \equiv 1 \bmod m, \tag{4.3}$$

where the solution $(3d', d')$ mod m should be regarded as "$(3/5, 1/5)$ mod m."

Case 3: $m \geq 2$ is arbitrary. We want to show $2x^2 + 7y^2 \equiv 1 \bmod m$ is solvable. If m is not divisible by 3 (but could be divisible by 5) then we can use (4.2). If m is not divisible by 5 (but could be divisible by 3) then we can use (4.3). What do we do if m is divisible by both 3 and 5? Use the Chinese remainder theorem!

Factor the biggest power of 3 from m: $m = 3^e M$ with $e \geq 1$ and $(3, M) = 1$. (Since m is divisible by 5, $M > 1$.)

- Since 3^e is not divisible by 5, (4.3) tells us that $\boxed{2(3d')^2 + 7d'^2 \equiv 1 \bmod 3^e}$ where $5d' \equiv 1 \bmod 3^e$. (The solution $(3d', d')$ mod 3^e is like "$(3/5, 1/5)$ mod 3^e.")
- Since M is not divisible by 3, (4.2) tells us that $\boxed{2d^2 + 7d^2 \equiv 1 \bmod M}$ where $3d \equiv 1 \bmod M$. (The solution (d, d) mod M is like "$(1/3, 1/3)$ mod M.")

In order to solve $2x^2 + 7y^2 \equiv 1 \bmod m$, we use integers x and y such that

$$x \equiv 3d' \bmod 3^e, \quad x \equiv d \bmod M, \qquad y \equiv d' \bmod 3^e, \quad y \equiv d \bmod M.$$

Such x and y exist by the Chinese remainder theorem. Then $2x^2 + 7y^2 \equiv 2(3d')^2 + 7d'^2 \equiv 1 \bmod 3^e$ and $2x^2 + 7y^2 \equiv 2d^2 + 7d^2 \equiv 1 \bmod M$, so $\boxed{2x^2 + 7y^2 \equiv 1 \bmod 3^e M}$.

For instance, take $m = 105 = 3 \cdot 5 \cdot 7$. Write $m = 3M$ where $M = 35$. The inverse of 5 mod 3 is $d' = 2$, so in place of $(3/5, 1/5)$ use $(3d', d') = (3 \cdot 2, 2) \bmod 3 = (0, 2) \bmod 3$. The inverse of 3 mod 35 is $d = 12$, so in place of $(1/3, 1/3)$ use $(d, d) = (12, 12) \bmod 35$.

Now we solve

$$x \equiv 0 \bmod 3, \quad x \equiv 12 \bmod 35, \qquad y \equiv 2 \bmod 3, \quad y \equiv 12 \bmod 35,$$

which turns out to be $x \equiv 12 \bmod 105$, $y \equiv 47 \bmod 105$, and indeed $2 \cdot 12^2 + 7 \cdot 47^2 \equiv 1 \bmod 105$.

To check that you understand how this example worked, use the solutions $(9/2, 1/2)$ and $(32/3, 5/3)$ of $x^2 - 37y^2 = 11$ to show you can solve $x^2 - 37y^2 \equiv 11 \bmod m$ for every $m \geq 2$.

(It can be shown by other arguments that $x^2 - 37y^2 = 11$ has no solution in $\mathbf{Z}$, which makes its solvability mod m for all m interesting!)

Theorem 4.17. *If the equation $ax^2 + by^2 = c$, where $a, b, c \in \mathbf{Z}$, has two rational solutions $(k/d, \ell/d)$ and $(k'/d', \ell'/d')$ where the denominators d and d' are relatively prime then the congruence $ax^2 + by^2 \equiv c \bmod m$ has a solution for every $m \geq 2$.*

Proof. Exercise, adapting the ideas of the preceding example. We can assume $d > 1$ and $d' > 1$, since if $d = 1$ or $d' = 1$ then one of the rational solutions is an integral solution and that will be a solution mod m directly for all m. □

We need the two rational solutions in the theorem to have relatively prime denominators in order to get a solution mod m for every m. Consider, for instance $x^2 - 41y^2 = 2$, which has the solution $(x, y) = (7/2, 1/2)$. This is enough to get a solution to $x^2 - 41y^2 \equiv 2 \bmod m$ for all odd $m > 1$ by inverting 2 modulo m, but without an additional rational solution having an odd denominator we don't get a solution modulo m for m an arbitrary power of 2. And indeed, there is no solution to $x^2 - 41y^2 \equiv 2 \bmod 2^e$ when $e \geq 2$ since there is no solution modulo 4.

Our final application of the Chinese remainder theorem is to an interpolation problem, to see that the scope of the Chinese remainder theorem goes beyond the setting of just the integers.

Given n points in the plane, $(a_1, b_1), \ldots, (a_n, b_n)$, with the a_i's distinct, we would like to find a polynomial $f(T)$ in $\mathbf{R}[T]$ whose graph passes through these points: $f(a_i) = b_i$ for $i = 1, 2, \ldots, n$. This task can be converted to a set of simultaneous congruences in $\mathbf{R}[T]$, which can be solved using the Chinese remainder theorem in $\mathbf{R}[T]$, not $\mathbf{Z}$. First let's state the Chinese remainder theorem for polynomials.

Theorem 4.18. *For $r \geq 2$, let $m_1(T), m_2(T), \ldots, m_r(T)$ be nonzero polynomials in $\mathbf{R}[T]$ which are pairwise relatively prime: $(m_i(T), m_j(T)) = 1$ for $i \neq j$. Then, for all polynomials $a_1(T), a_2(T), \ldots, a_r(T)$ in $\mathbf{R}[T]$, the system of congruences*

$$f(T) \equiv a_1(T) \bmod m_1(T), \quad f(T) \equiv a_2(T) \bmod m_2(T), \quad \ldots, \quad f(T) \equiv a_r(T) \bmod m_r(T),$$

has a solution $f(T)$ in $\mathbf{R}[T]$, and this solution is unique modulo $m_1(T)m_2(T)\cdots m_r(T)$.

The proof of this is identical to that of the Chinese remainder theorem for $\mathbf{Z}$, so we leave it to the reader as an exercise.

Theorem 4.19. *In $\mathbf{R}$, pick two lists of n numbers $a_1, a_2, \ldots, a_n$ and $b_1, b_2, \ldots, b_n$ where the a_i's are distinct. There is a unique polynomial $f(T)$ of degree $< n$ in $\mathbf{R}[T]$, possibly 0, such that $f(a_i) = b_i$ for all i.*

Proof. To say $f(a_i) = b_i$ is the same as $f(T) \equiv b_i \bmod T - a_i$ (why?). Consider the system of congruences

$$f(T) \equiv b_1 \bmod T - a_1, f(T) \equiv b_2 \bmod T - a_2, \ldots, f(T) \equiv b_n \bmod T - a_n$$

for an unknown $f(T)$ in $\mathbf{R}[T]$. Since the a_i's are *distinct*, the polynomials $T - a_1, \ldots, T - a_n$ are pairwise relatively prime in $\mathbf{R}[T]$. Therefore, by the Chinese remainder theorem in $\mathbf{R}[T]$, there is an $f(T)$ in $\mathbf{R}[T]$ satisfying all of the above congruences. It follows that $f(a_i) = b_i$ for all i.

We have no initial control over $\deg f$ for the common solution f. However, since we can adjust $f(T)$ modulo $(T - a_1)\cdots(T - a_n)$ without changing the congruence conditions, we

can replace $f(T)$ with its remainder under division by $(T-a_1)\cdots(T-a_n)$, which has degree n. Then $\deg f < n$ with $f(a_i) = b_i$ for all i.

We have shown a desired $f(T)$ exists. To see it is unique, suppose $f_1(T)$ and $f_2(T)$ both have degree less than n and satisfy

$$f(T) \equiv b_1 \bmod T - a_1, f(T) \equiv b_2 \bmod T - a_2, \ldots, f(T) \equiv b_n \bmod T - a_n.$$

Then, by the uniqueness in the Chinese remainder theorem, we have

$$f_1(T) \equiv f_2(T) \bmod (T - a_1)\cdots(T - a_n).$$

Since $f_1(T)$ and $f_2(T)$ have degree less than n, this congruence modulo a polynomial of degree n implies $f_1(T) = f_2(T)$ in $\mathbf{R}[T]$. □

The fact that polynomial interpolation is identical to solving a system of polynomial congruences (with linear moduli) suggests that we should think about solving a system of integer congruences as *arithmetic* interpolation.

There is nothing essential about $\mathbf{R}$ in Theorem 4.19 except that it's a field. The Chinese remainder theorem goes through for $F[T]$ with F a general field, not just $\mathbf{R}$, and Theorem 4.19 carries over to a general field:

Theorem 4.20. *Let F be a field. For n distinct numbers $a_1, a_2, \ldots, a_n$ in F and arbitrary numbers $b_1, b_2, \ldots, b_n$ in F, which may have repetitions, there is a unique polynomial $f(T)$ of degree $< n$ in $F[T]$, possibly 0, such that $f(a_i) = b_i$ for all i.*

The proof is identical to that of Theorem 4.19.

References

[1] R. J. Sullivan, "Radar Foundations for Imaging and Advanced Concepts," Scitech Publishing, 2004.